

PRINCIPLES & PROCESSES OF PLANT RESPIRATION

Complete Comprehensive Solutions and Academic Study Guide (Class 7 Biology)

This document provides authoritative, fully-solved answers for the review questions and experimental exercises from the curriculum textbook on **Photosynthesis and Respiration in Plants**. The solutions are rigorously grounded in the principles of plant physiology, including carbon assimilation mechanisms, carbohydrate storage, aerobic and anaerobic pathways, vascular transport, and gaseous exchange dynamics. This manual serves as a complete study resource, incorporating structured comparison tables, detailed anatomical explanations of stomatal behavior, and comprehensive analyses of classic laboratory experiments.

SECTION 1: OBJECTIVE TYPE QUESTIONS

1. Multiple Choice Questions (Select the most appropriate alternative)

(i) Carbohydrates are stored by plants in the form of:

- (a) Starch (b) Proteins (c) Fats (d) Glucose

Explanation: Glucose produced during photosynthesis is converted into insoluble starch for temporary storage in the leaves and long-term storage in storage organs.

(ii) Stomata are present on the surface of:

- (a) Leaves (b) Roots (c) Stem (d) Flower petals

Explanation: Stomata are minute microscopic pores found predominantly on the epidermal surface of leaves, particularly on the lower epidermis to regulate transpiration.

(iii) Which one of the following is the immediate end product of photosynthesis?

- (a) Fructose (b) Glucose (c) Cellulose (d) Lactose

Explanation: Glucose is the primary simple carbohydrate synthesized directly in the mesophyll cells during the chemical reactions of photosynthesis.

(iv) The process by which green plants prepare their own food is known as:

- (a) Photorespiration (b) Photolysis (c) Photosynthesis (d) None of the above

Explanation: Photosynthesis is the light-driven metabolic process where autotrophs combine carbon dioxide and water to synthesize glucose, releasing oxygen as a by-product.

(v) In which of the following cellular structures does photosynthesis occur?

- (a) Chromoplast (b) Chromosome (c) Chloroplast (d) Leukoplast

Explanation: Chloroplasts are specialized double-membrane organelles containing chlorophyll that capture solar energy to perform photosynthesis.

(vi) Which of the following forms of carbohydrates is broken down during respiration?

- (a) Cellulose **(b) Glucose** (c) Starch (d) Sucrose

Explanation: Glucose serves as the primary respiratory substrate that is oxidized inside living cells to release chemical energy in the form of ATP.

(vii) Which of the following is an immediate product of photosynthesis?

- (a) Glucose** (b) Starch (c) Sucrose (d) Cellulose

Explanation: Glucose is the immediate simple sugar formed, which is subsequently converted into sucrose for transport, or polymerized into starch for storage.

(viii) Which of the following is an end product of anaerobic respiration in plants?

- (a) Ethane (b) Ethylene **(c) Ethanol** (d) Lactic acid

Explanation: In anaerobic plant respiration (fermentation), glucose is partially broken down in the absence of oxygen to yield ethanol (ethyl alcohol), carbon dioxide, and 2 ATP molecules.

(ix) A glowing matchstick starts burning in the presence of:

- (a) Carbon dioxide gas (b) Nitrogen gas (c) Carbon monoxide gas **(d) Oxygen gas**

Explanation: Oxygen is a strong supporter of combustion, which causes a glowing splinter or matchstick to rekindle and burn brightly.

(x) Which of the following chemicals is used to identify the presence of starch?

- (a) Lime water (b) Potassium hydroxide **(c) Iodine solution** (d) Methylated alcohol

Explanation: Iodine solution is the standard chemical indicator for starch, reacting with starch to produce a characteristic deep blue-black color complex.

2. Leaf vs. Factory Comparison (Analogy Match)

Comparing the structural and functional elements of a green leaf to a manufacturing factory provides a clear, metaphorical understanding of plant physiology. The matching elements from Column A and Column B are compiled below:

Leaf Element (Column A)	Factory Equivalent (Column B)	Physiological Mechanism & Role
(i) Cells in the leaf	(f) Work room	The individual cells (especially mesophyll) are the physical workspaces where all biochemical reactions take place.
(ii) Chloroplast	(c) Machinery	The chloroplasts are the functional organelles (machines) containing the apparatus to carry out synthesis.
(iii) Sunlight	(b) Power	Solar energy provides the thermodynamic driving force (electrical/mechanical power) that fuels the endergonic reactions.
(iv) Oxygen and water	(e) By-product	Oxygen and excess water vapour are released as outputs during the synthesis, which are not the primary target products.
(v) Carbon dioxide and water	(a) Raw materials	CO ₂ and H ₂ O are the inorganic substrates (feedstock) taken in to synthesize organic molecules.
(vi) Glucose	(d) End product	Glucose is the primary manufactured carbohydrate (finished goods) produced at the completion of the assembly process.

3. True or False Statements

(i) Statement: Green plants prepare their food by using two raw materials, oxygen and water.

Answer: FALSE — Green plants use carbon dioxide (CO₂) and water (H₂O) as raw materials for photosynthesis. Oxygen (O₂) is not a raw material; it is a gaseous by-product of photolysis that is released into the atmosphere.

(ii) Statement: The chlorophyll enables the plants to use light energy.

Answer: TRUE — Chlorophyll is a specialized photosynthetic pigment located in the thylakoid membranes of chloroplasts that absorbs photons of light, initiating the conversion of solar energy into chemical energy.

(iii) Statement: The free oxygen in the atmospheric air is the result of photosynthesis.

Answer: TRUE — Virtually all molecular oxygen (O₂) in the Earth's atmosphere is generated by oxygenic photosynthesis, where water molecules are split (photolysis) by green plants, algae, and cyanobacteria.

(iv) Statement: Photosynthesis occurs only in chlorophyll-containing parts of the plant.

Answer: TRUE — The light reaction of photosynthesis strictly requires chlorophyll to capture solar energy. Therefore, non-green parts (such as roots, woody stems, and white patches of variegated leaves) cannot perform photosynthesis.

4. Gas Exchange in Bright Sunlight

(i) Gas taken in: Carbon dioxide (CO₂)

(ii) Gas given out: Oxygen (O₂)

Physiological Explanation: In bright sunlight, green leaves perform both photosynthesis and respiration. However, the rate of photosynthesis is significantly higher (typically 10-20 times) than the rate of respiration. Consequently, the volume of CO₂ absorbed for photosynthesis vastly exceeds the volume of CO₂ released by respiration. Likewise, the volume of O₂ released as a by-product of photosynthesis is much greater than the O₂ consumed in respiration, leading to a net intake of carbon dioxide and net release of oxygen.

5. Name the Following

(i) Description: The cells which surround the openings of stomata

Term: Guard cells — *Bean-shaped epidermal cells that regulate the opening and closing of the stomatal pore via changes in turgor pressure.*

(ii) Description: The chemical energy stored in the cells

Term: Adenosine Triphosphate (ATP) / Glucose — *ATP is the immediate energy currency of the cell, synthesized during respiration by the oxidation of glucose molecules.*

(iii) Description: The pigment which traps solar energy for the process of photosynthesis

Term: Chlorophyll — *A green photoreceptive pigment located in the chloroplasts that selectively absorbs light in the blue and red regions.*

(iv) Description: A non-chlorophyllous, unicellular plant that respire anaerobically

Term: Yeast — *A unicellular fungus that lacks chlorophyll and carries out anaerobic respiration (alcoholic fermentation) to release energy.*

(v) Description: The process which maintains the coolness of the plant body

Term: Transpiration — *The loss of water in the form of water vapour from the aerial parts of the plant (primarily stomata), producing an evaporative cooling effect.*

6. Note the relationship and complete the analogy

(i) Analogy: Photosynthesis : Day time :: Respiration : Day and night (or Continuous)

Reason: Photosynthesis is light-dependent and occurs only during the daytime, whereas respiration is a fundamental metabolic process that must occur continuously (both day and night) in all living cells to maintain life.

(ii) Analogy: Photo : light :: Synthesis : Building up (or Chemical combination / Preparation)

Reason: The term 'Photosynthesis' is derived from Greek roots where 'photo' means light and 'synthesis' refers to the process of putting together or building up organic molecules from inorganic substrates.

(iii) Analogy: Xylem : Water and minerals :: Phloem : Prepared food (or Sucrose / Organic solutes)

Reason: Xylem is the vascular tissue responsible for the unidirectional conduction of water and dissolved minerals from the roots to the leaves. Phloem is responsible for the multidirectional translocation of manufactured organic food from the leaves to other parts of the plant.

(iv) Analogy: Palisade cells : Barrel-shaped :: Guard cells : Bean-shaped (or Kidney-shaped / Crescent)

Reason: Palisade cells in the leaf mesophyll are elongated, tightly packed, barrel-shaped columns. In contrast, epidermal guard cells are specialized crescent-shaped or bean-shaped structures that curve to enclose the stomatal pore.

(v) Analogy: Photosynthesis : Chloroplast :: Respiration : Mitochondria (and Cytoplasm)

Reason: Photosynthesis occurs within the specialized chloroplast organelles. Respiration begins in the cytoplasm (glycolysis) and is completed within the mitochondria (aerobic Krebs cycle and electron transport chain) of living cells.

7. Fill in the Blanks

(i) Plants release oxygen gas as a by-product during photosynthesis.

Scientific context: Oxygen is produced via the photolysis of water during the light reaction of photosynthesis and diffuses out of the leaves.

(ii) Plants use oxygen gas in the process of respiration during day time.

Scientific context: Aerobic respiration occurs continuously in plants, requiring oxygen to oxidize glucose, regardless of whether it is day or night.

(iii) Carbon dioxide gas turns lime water milky.

Scientific context: Carbon dioxide reacts with calcium hydroxide $[\text{Ca}(\text{OH})_2]$ in lime water to form an insoluble white precipitate of calcium carbonate $[\text{CaCO}_3]$.

(iv) Methylated alcohol is used to decolourise the leaf.

Scientific context: Boiling a leaf in methylated alcohol extracts and dissolves chlorophyll, stripping the leaf of its green pigment so that starch-iodine color changes can be observed.

(v) Iodine solution turns starch into blue-black colour.

Scientific context: The iodine-potassium iodide complex intercalates into the helical structure of amylose (starch), forming a characteristic intense blue-black complex.

SECTION 2: SHORT ANSWER QUESTIONS

Q1 (i): Name two main modes of nutrition in the organisms.

Answer: The two primary modes of nutrition are:

1. **Autotrophic nutrition:** The mode of nutrition where organisms synthesize their own organic food from simple inorganic raw materials (carbon dioxide and water) using light or chemical energy (e.g., green plants, autotrophic bacteria).
2. **Heterotrophic nutrition:** The mode of nutrition where organisms cannot manufacture their own food and must obtain organic carbon by consuming other organisms (e.g., animals, fungi, non-chlorophyllous plants).

Q1 (ii): Name four essential requirements for the process of photosynthesis.

Answer: The four absolute requirements for a plant to carry out photosynthesis are:

1. **Chlorophyll:** The green photoreceptive pigment that absorbs light energy.
2. **Sunlight:** The solar radiant energy source that drives the endergonic chemical reactions.
3. **Carbon dioxide (CO₂):** Taken from the atmospheric air, providing the carbon and oxygen source for glucose synthesis.
4. **Water (H₂O):** Absorbed from the soil by roots, serving as the hydrogen source and the donor of electrons (via photolysis).

Q1 (iii): Name two categories of cells in the mesophyll of leaves.

Answer: The mesophyll tissue of a leaf is differentiated into:

1. **Palisade mesophyll cells:** Elongated, column-like, barrel-shaped cells arranged vertically in tight rows just beneath the upper epidermis, packed with chloroplasts to maximize light absorption.
2. **Spongy mesophyll cells:** Irregularly shaped, loosely arranged cells located below the palisade layer, with extensive intercellular air spaces that facilitate the diffusion of gases (CO₂ and O₂) to and from the stomata.

Q1 (iv): Name two types of tissues of vascular system.

Answer: The plant vascular transport system is composed of:

1. **Xylem:** A complex conducting tissue composed of tracheids, vessels, xylem parenchyma, and xylem fibres, specialized for the conduction of water and dissolved mineral nutrients from roots upwards to the aerial parts.
2. **Phloem:** A complex conducting tissue composed of sieve tubes, companion cells, phloem parenchyma, and phloem fibres, specialized for the translocation of manufactured organic food (sucrose) in solution from source organs (leaves) to sink organs (roots, shoots, developing fruits).

Q1 (v): Name two types of respiration in the organisms.

Answer: The two main types of cellular respiration are:

1. **Aerobic respiration:** The metabolic breakdown of glucose in the presence of oxygen, resulting in complete oxidation to carbon dioxide and water, releasing a high yield of energy (38 ATP molecules per glucose molecule).
2. **Anaerobic respiration:** The partial breakdown of glucose in the complete absence of oxygen, yielding ethanol and carbon dioxide (in plants/yeast) or lactic acid (in animal muscle cells), releasing a low yield of energy (2 ATP molecules per glucose molecule).

Q2: Define the following terms: (i) Autotrophs, (ii) Photosynthesis, (iii) Respiration, (iv) Vascular system, (v) Stomata.

Answer: (i) Autotrophs: Organisms that can synthesize their own organic molecules (food) from simple inorganic substances (like carbon dioxide and water) using light or chemical energy (e.g., green plants).

(ii) Photosynthesis: The biochemical process by which green plants containing chlorophyll capture light energy and utilize it to combine carbon dioxide and water into carbohydrate (glucose) molecules, releasing oxygen as a by-product.

(iii) Respiration: The metabolic process occurring within all living cells where complex organic molecules (glucose) are oxidatively broken down to release energy in the form of ATP, with carbon dioxide and water formed as waste products.

(iv) Vascular system: The specialized internal transport network of plants consisting of xylem and phloem vessels, responsible for moving water, mineral ions, and organic nutrients throughout the plant.

(v) Stomata: Tiny, microscopic pores present in the epidermal layer of leaves and young stems, bounded by a pair of guard cells, which regulate transpiration and the exchange of carbon dioxide and oxygen between the plant and the atmosphere.

Q3: Why do leaves generally look green?

Answer: Leaves appear green because they contain chloroplasts in their mesophyll cells, which are rich in the green pigment **chlorophyll**. Chlorophyll molecules absorb light energy highly in the blue (400-450 nm) and red (640-680 nm) regions of the solar spectrum, but they reflect green light (500-550 nm). The reflected green light is detected by our eyes, making the foliage appear green.

Q4: What is the source of energy for photosynthesis?

Answer: The primary source of energy for photosynthesis is **sunlight** (solar radiant energy). Chlorophyll molecules in the chloroplasts absorb photons of light, and this absorbed light energy is used to photolyse water and synthesize chemical energy (ATP and NADPH), which subsequently fuels the dark reactions to fix carbon dioxide into glucose.

Q5: Do the plants respire all day and night or only during the day? Give reasons.

Answer: Plants **respire continuously, all day and night**. Respiration is a foundational metabolic process required by all living cells to produce ATP, which powers cellular maintenance, active transport of minerals, cell division, and other life-sustaining functions. Unlike photosynthesis, which is light-dependent and occurs only during the daytime, respiration is light-independent and must occur non-stop. If cellular respiration ceases, even briefly, the cells will starve of ATP and the plant will die.

Q6: What happens to the energy liberated during respiration?

Answer: When glucose molecules are oxidized during cellular respiration, a small portion of the chemical energy is liberated as thermal energy (heat). However, the major portion is captured and stored in the chemical bonds of **Adenosine Triphosphate (ATP)** molecules (aerobic respiration yields 38 ATP per glucose). These ATP molecules serve as the universal energy currency, diffusing to wherever energy-requiring work is performed in the cell.

Q7: What is the role of chlorophyll in photosynthesis?

Answer: Chlorophyll acts as a vital photosynthetic photoreceptor. Located within the thylakoid membranes of chloroplasts, its roles include:

1. **Absorption of Light:** It selectively traps solar energy (photons).
2. **Energy Conversion:** It converts the captured light energy into high-energy chemical bonds (ATP and NADPH) through the light reaction pathways.
3. **Photolysis of Water:** It facilitates the splitting of water molecules into hydrogen ions, electrons, and oxygen, initiating electron transport.

Q8: Do plants need oxygen? If so, what is its source?

Answer: Yes, plants require oxygen (O_2) for aerobic respiration to completely oxidize glucose and release the maximum yield of ATP. Their sources of oxygen differ by time of day:

1. **During the Day:** The primary source is the oxygen produced internally as a by-product of active photosynthesis. This oxygen is directly channeled from the chloroplasts to the mitochondria.
2. **During the Night:** Since photosynthesis is inactive, plants must absorb oxygen from the surrounding environment. It diffuses into the plant from the atmosphere through the stomata (leaves), lenticels (woody stems), and root hairs (soil air spaces).

SECTION 3: LONG ANSWER QUESTIONS

1. Distinguish between the following pairs on the basis of the indicated criteria

(i) Photosynthesis vs. Respiration

Key Feature	Photosynthesis	Respiration
Raw Materials	Carbon dioxide (CO_2) and Water (H_2O)	Glucose ($C_6H_{12}O_6$) and Oxygen (O_2)
Energy Exchange	Endergonic (stores light energy as chemical energy in glucose)	Exergonic (releases chemical energy from glucose as ATP and heat)
Site of Process	Chloroplasts (exclusively in green cells containing chlorophyll)	Cytoplasm and Mitochondria (in all living cells of the organism)
Occurrence Time	Occurs only during daytime in the presence of sunlight	Occurs continuously, both day and night

(ii) Aerobic Respiration vs. Anaerobic Respiration (in Plants)

Key Feature	Aerobic Respiration	Anaerobic Respiration (in Plants)
End Products	Carbon dioxide (CO_2), Water (H_2O), and energy	Ethanol (C_2H_5OH), Carbon dioxide (CO_2), and energy
Oxygen Requirement	Strictly utilizes molecular oxygen (O_2) for complete oxidation	Does not utilize oxygen; organic molecules are partially broken down
Energy Yield	High yield: 38 ATP molecules per glucose molecule	Low yield: 2 ATP molecules per glucose molecule

Key Feature	Aerobic Respiration	Anaerobic Respiration (in Plants)
Substrate breakdown	Glucose is completely broken down to CO_2 and H_2O	Glucose is partially broken down to ethanol and CO_2

(iii) Green Leaves vs. Variegated Leaves

Key Feature	Green Leaves	Variegated Leaves
Examples	Mango, Peepal, Rose, Grass, Hibiscus	Coleus, Croton, Geranium, Variegated Ivy
Chlorophyll Distribution	Chlorophyll is uniformly distributed across the entire leaf lamina.	Chlorophyll is confined to specific green patches; white/yellow areas lack chlorophyll.
Photosynthetic Capacity	The entire surface of the leaf is photosynthetic and manufactures starch.	Only the green, chlorophyll-containing patches can perform photosynthesis and store starch.

(iv) Xylem vs. Phloem Tissues

Key Feature	Xylem Tissue	Phloem Tissue
Function	Conduction of water and dissolved mineral ions from roots upward to leaves.	Translocation of manufactured organic food (sucrose) from leaves to all parts of the plant.
Direction of Flow	Unidirectional flow (strictly upwards from roots to shoots).	Multidirectional flow (upwards to growing tips and downwards to roots).
Structural Components	Composed of vessels, tracheids, fibres (dead cells) and parenchyma (living).	Composed of sieve tubes, companion cells, parenchyma (living) and fibres (dead).
Conduction Mechanism	Passive transport driven by transpirational pull and root pressure.	Active transport (translocation) driven by osmotic pressure gradients (loading/unloading).

(v) Glucose vs. Starch

Key Feature	Glucose	Starch
Solubility in Water	Highly soluble in water. Dissolves easily in cellular fluids.	Insoluble in water. Forms a colloidal suspension; does not affect osmotic potential.
Function / Role	Immediate substrate for respiration and the direct product of carbon synthesis.	Form of storage carbohydrate reserved in plastids for future energy demands.
Molecular Structure	A simple, single-ring monomeric sugar (monosaccharide: $C_6H_{12}O_6$).	A complex macromolecular polymer of glucose monomers (polysaccharide).

2. Diagram-Based Study: Opened and Closed Stomata

(i) Labeling of Key Anatomical Parts:

- *Guard Cells*: The two kidney/bean-shaped epidermal cells containing chloroplasts.
- *Stoma (Stomatal Pore)*: The central microscopic aperture enclosed by the guard cells.
- *Nucleus*: The distinct round organelle in each guard cell coordinating cellular functions.
- *Inner Wall*: The thick, inelastic, elastic-resistant cell wall directly facing the stomatal pore.
- *Outer Wall*: The thin, highly elastic, stretchable cell wall facing away from the pore.
- *Chloroplasts*: Small green lens-shaped plastids scattered in the cytoplasm of the guard cells.

(ii) Condition of Guard Cells during Open and Closed States:

- *Open Stoma (Turgid Condition)*: During the day, solute concentration inside guard cells rises, causing water to enter them from surrounding epidermal cells via endosmosis. The guard cells swell and become **turgid**. Because the outer wall is thin and elastic, it bulges outwards, pulling the thick, inelastic inner wall along with it. This outward bowing creates the aperture of the stoma.
- *Closed Stoma (Flaccid Condition)*: At night (or during drought), water leaves the guard cells via exosmosis. The cells lose their turgidity and become **flaccid**. The elastic tension is released, and the thick inner walls return to their straight, relaxed positions, sealing the stomatal pore.

(iii) Specific Location in the Plant Body:

Stomata are specialized structures located in the **epidermal layer of leaves** and young, green herbaceous stems. In terrestrial dicotyledonous plants, they are predominantly situated on the **lower epidermis** (hypostomatic leaves) to restrict excessive transpirational water loss while allowing gas exchange.

(iv) Three Important Processes in which Stomata are Involved:

1. *Photosynthesis*: Serves as the pathway for the intake of carbon dioxide (CO_2) and the release of oxygen (O_2) during daytime.
2. *Respiration*: Facilitates the inhalation of oxygen (O_2) and the exhalation of carbon dioxide (CO_2) for cellular metabolism.
3. *Transpiration*: Regulates the loss of water vapour from the plant's interior, creating a pulling force (transpiration stream) for water/mineral transport and cooling the leaf surface.

(v) State during Day and Night:

Stomata are generally **open during the day** (when light is present to drive photosynthesis, necessitating CO_2 entry) and **closed during the night** (when photosynthesis is halted, allowing the plant to conserve water by stopping transpiration).

3. Figure-Based Study: Photosynthesis Scheme

The process represented in the schematic diagram (Sunlight + Green leaves + CO_2 + Water = Sugar + Oxygen) is **Photosynthesis**.

(i) Identification & Scientific Definition:

Photosynthesis is the fundamental photochemical process by which autotrophic green plants utilize light energy absorbed by chlorophyll to synthesize organic food (simple sugars like glucose) from inorganic raw materials such as carbon dioxide and water, releasing oxygen as a gaseous by-product.

(ii) Balanced Chemical Equation:

The complete balanced chemical reaction for oxygenic photosynthesis is represented as:



(iii) Cellular Organelle and Pigment:

- *Organelle*: The process takes place exclusively within the double-membrane-bound **Chloroplast** organelles of plant cells (mainly in leaf mesophyll).
- *Photosynthetic Pigment*: The chemical pigment responsible for trapping light photons is **Chlorophyll**, which resides in the thylakoid membranes of chloroplasts.

(iv) Two Gaseous By-Products:

1. **Oxygen (O_2)**: Released from the splitting of water molecules (photolysis) during the light reactions. It is the primary metabolic by-product.
2. **Water Vapour (H_2O)**: Though water is a reactant, moisture is also lost during stomatal opening, making water vapour a secondary physical product via transpiration.

(v) Significance of Photosynthesis (Two Major Points):

1. *Primary Source of Food (Energy Base)*: Photosynthesis is the only natural biological mechanism that fixes radiant solar energy into chemical energy. It synthesizes organic food molecules that form the foundation of food webs, supporting all herbivorous and carnivorous life on Earth.
2. *Oxygenation and Gas Regulation*: The process releases molecular oxygen, which is indispensable for the survival of all aerobic organisms. It also acts as the primary carbon sink, absorbing huge quantities of CO_2 from the atmosphere, maintaining a stable atmospheric gas balance.

4. Experimental Study: Demonstration of Oxygen Production

(i) Aim of the Experiment:

To experimentally demonstrate that oxygen (O₂) is liberated as a gaseous by-product during the process of photosynthesis in green plants.

(ii) Suitable Plant and its Habitat:

The suitable plant used is **Hydrilla** (or Elodea). Its habitat is an **aquatic, freshwater habitat** (ponds and slow-moving streams). Aquatic plants are selected because they release gaseous products as visible bubbles under water, which can easily displace water in an inverted container.

(iii) Importance of the Gas Formed for Us:

The gas formed is **oxygen**. This gas is absolutely essential for human beings and all aerobic organisms because it is the terminal electron acceptor in **cellular respiration**. It is required to break down food inside our mitochondria to produce ATP energy, which keeps us alive.

(iv) Chemical/Physical Confirmation of the Gas:

To confirm the presence of oxygen, remove the inverted test tube from the water by placing a thumb over its mouth to prevent gas escape. Turn the tube upright, remove the thumb, and immediately introduce a **glowing matchstick or a glowing wooden splinter** near the mouth. If the gas is oxygen, the glowing matchstick will instantly **re-kindle and burst into a bright, active flame**, confirming oxygen's property as a strong supporter of combustion.

(v) Explanation of the 'Initial Stage' of the Experiment:

In the 'initial stage' of the experimental set-up, the beaker is filled with fresh pond water, and Hydrilla twigs are placed under an inverted glass funnel. A glass test tube is **completely filled with water**, sealed with a finger, and carefully inverted over the stem of the funnel under water, ensuring that **no air bubbles or gas space exist at the closed top of the test tube**. The entire apparatus is then placed in bright sunlight. At this initial juncture, no gas has been generated, so the test tube remains completely filled with water.

5. Balanced Chemical Equations for Key Processes

The complete chemical equations representing the key metabolic and physical processes in plant and animal life are formatted below:

(i) Photosynthesis

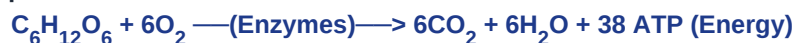
Equation:



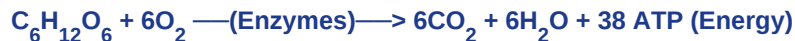
Description: Six molecules of carbon dioxide react with six molecules of water in the presence of solar energy trapped by chlorophyll to synthesize one molecule of glucose and release six molecules of oxygen gas.

(ii) Aerobic Respiration in Animals

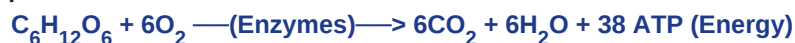
Equation:



Description: One molecule of glucose is completely oxidized by six molecules of oxygen in the cytoplasm and mitochondria, yielding carbon dioxide, water, and 38 ATP molecules of biochemical energy.

(iii) Aerobic Respiration in Plants during Daytime**Equation:**

Description: Respiration occurs continuously in plant cells during the day, utilizing oxygen (mostly produced by photosynthesis) to oxidize glucose and produce CO₂, water, and ATP energy.

(iv) Aerobic Respiration in Plants during Night**Equation:**

Description: At night, since photosynthesis is inactive, plant cells absorb environmental oxygen to drive the exact same aerobic respiratory pathway, producing CO₂, water, and ATP energy.

(v) Anaerobic Respiration in Plants**Equation:**

Description: In the absence of oxygen, glucose is partially oxidized (fermented) in the cytoplasm to yield two molecules of ethanol, two molecules of carbon dioxide, and a minor energy yield of 2 ATP molecules.